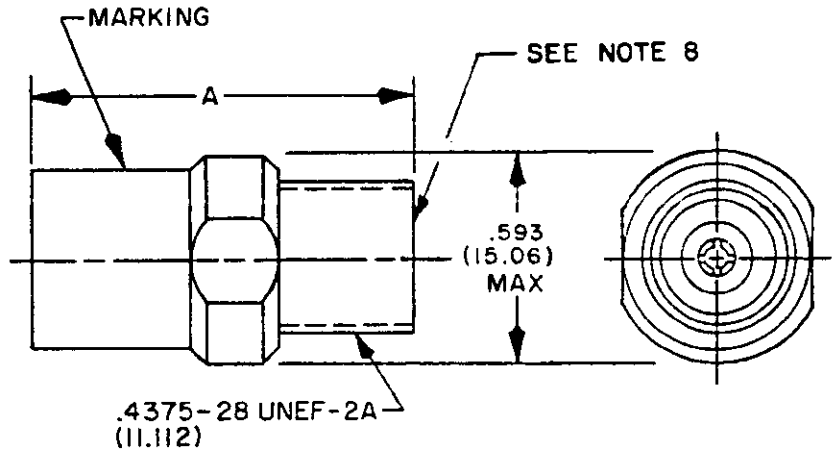
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES TNC (CABLED), SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



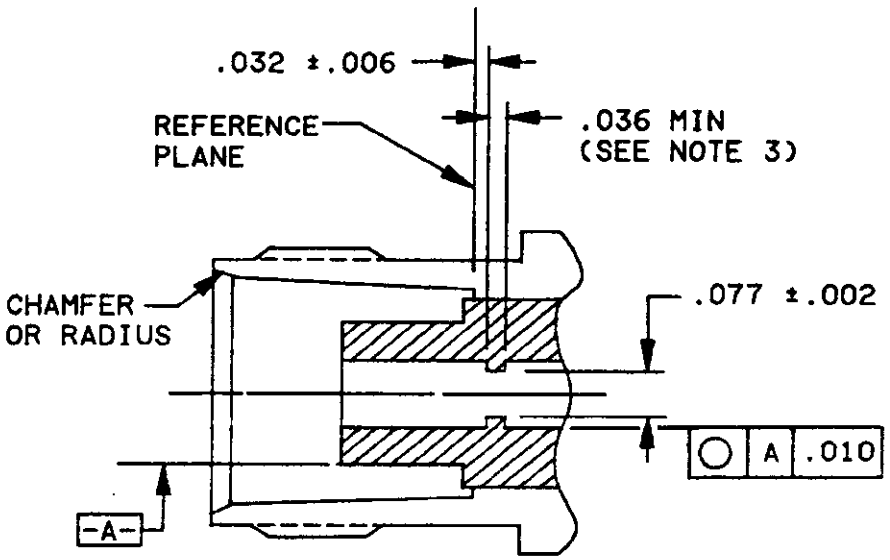
MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE IV

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see table I and III.
4. Dimension .593 (15.06 mm) is the largest overall diameter of the connector.
5. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
8. Series TNC, socket contact interface in accordance with MIL-STD-348.
9. Metrics equivalents are in parentheses.

FIGURE 1. General configuration.

AMSC N/A FSC 5935

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CATEGORY D

Inches	mm
.002	0.05
.006	0.15
.010	0.25
.032	0.81
.036	0.91
.077	1.96
.4375	11.112

NOTES:
1. Metric equivalents are given for general information only.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Chamfer is optional.
4. Dimensions are in inches.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/#	Dimensions	Inches (millimeters) maximum
Category A - Field serviceable (no special tools required)			
0101 3/	028-RG058 084-RG223* 128-RG400 060-RG142@	A	1.250 (31.75)
0002 3/	29-RG59*Δ 030-RG062Δ 90-RG71Δ 97-RG210Δ		
0103 3/	054-RG122*		
0104 3/	111-RG303@*		
0117 3/	110-RG302*@Δ		
0018 3/	113-RG316*@ 119-RG174		
Category C - Field replaceable (MIL-C-22520 crimp tool) 2/ See footnote symbol next to applicable cable for crimp die			
0010 3/	028-RG058*~ 111-RG303@~	A	1.500 (38.10)
0011 3/	084-RG223*~ 128-RG400~ 060-RG142@~		
0012 3/	29-RG59*Δ> 030-RG062Δ> 97-RG210Δ>		
0013 3/	054-RG122*&		
0014 3/	90-RG71*Δ>		
0021 3/	110-RG302*Δ@>		
0022 3/	113-RG316@*Σ 119-RG174Σ		
0023 3/	95-RG180&		

See footnotes at end of table.

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TABLE I. Dash numbers, cross reference, and dimensions - Continued.

Dash number 1/	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
Category D - Field replaceable - Defined piece parts 2/4/5/			
0501 3/	054-RG122*	A	1.234 (31.35)
0502 3/	95-RG180*Δ		
0503 3/	060-RG142* 128-RG400		
0504 3/	028-RG58*		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

3/ These connectors have captivated center contacts.

4/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

5/ Not to be used in Army equipment.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

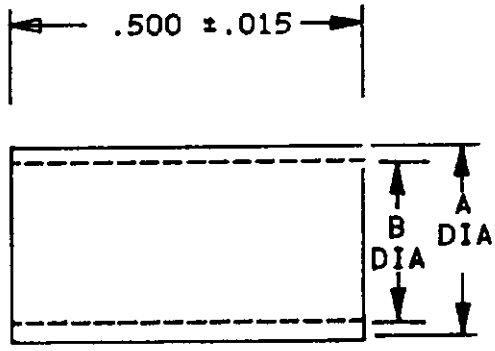
Σ M22520/5-35 Closure B or M22520/5-03 Closure A

& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

- M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

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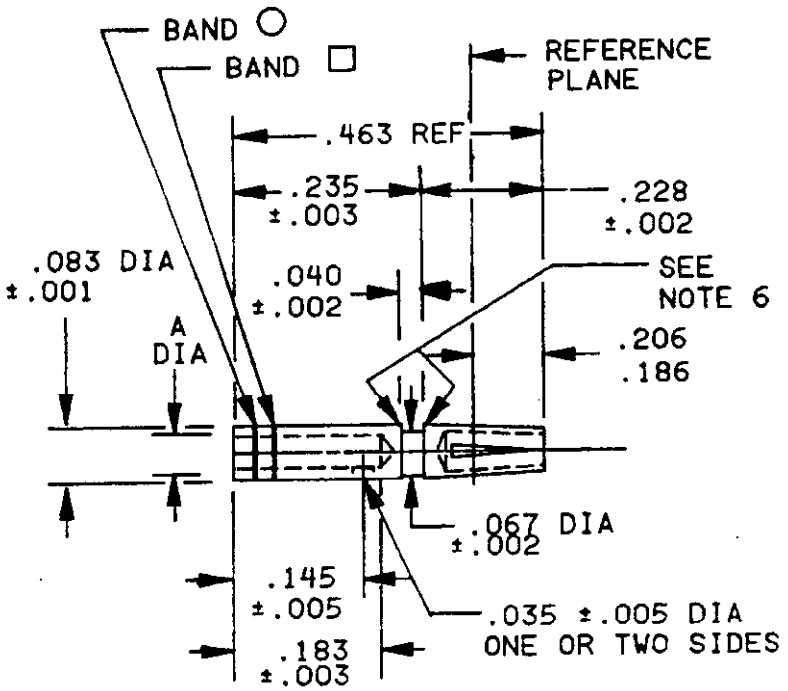
CRIMP FERRULE

Dash no.	Ferrule no. 2/	A +.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner M22520/5-
0501 0502	27-50	.212	.175	M22520/5-01	05, 41 Closure B or 9 Closure A
0503	27-51	.250	.220		05, 11, 57 Closure A or 19
0504	27-52	.245	.206		Closure B

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.015	0.38
.017	0.43
.033	0.84
.035	0.89
.040	1.02
.043	1.09
.067	1.70
.083	2.11
.145	3.68
.175	4.45
.183	4.65
.186	4.72
.206	5.23
.212	5.38
.220	5.59
.228	5.79
.235	5.97
.250	6.35
.463	11.76
.500	12.70

CENTER CONTACT

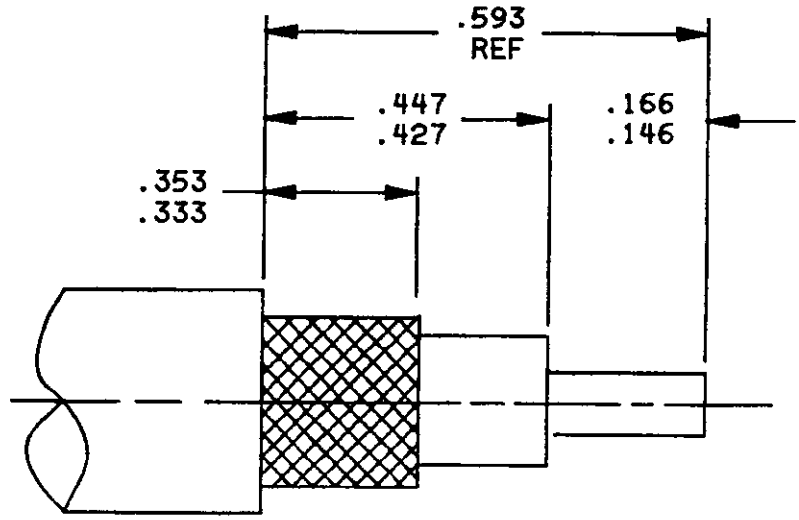
Dash no.	Contact no. 2/	A +.001 -.002	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501	27-12	.033	M22520/1-01	M22520/1-12	10 lbs	Orn	Green
0502	27-11	.017			6 lbs	Blue	
0503 0504	27-10	.043	M22520/1-01	M22520/1-12	20 lbs	Red	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 (0.08 mm) maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.146	3.71
.166	4.22
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.593	15.06

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions:

Coupling torque: 4 to 6 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: .057 diameter, minimum (nonclosed entry contacts only).

Insertion depth: .125, minimum.

Number of insertions: One.

Insertion force test: Steel test pin diameter .054, minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .052, maximum.

Withdrawal force: 2 ounces, minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition 3.

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Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.; 1.30, maximum.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.02+.003 F (F in GHz).

Item 16: VSWR shall be less than 1.02+.003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06+.007 F (F in GHz).

Group B inspection: VSWR shall be less than 1.05+.005 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.10.

Connector durability: 500 cycles, *minimum* at 12 cycles per minute, *maximum*. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms, minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition G. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 375 volts rms, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 40 pounds, minimum.

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Crimp assemblies:

10 pounds, minimum for cables .155 - .189 OD.

20 pounds, minimum for cables .190 - .229 OD.

30 pounds, minimum for cables .230 - .249 OD.

40 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:

.18 dB maximum tested at 9 GHz.

.06 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/27 (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
I	27-0101 27-0103 27-0104 27-0018 28-0101 28-0103 28-0104 28-0018 29-0101 29-0103 29-0104 29-0018	27-0101 27-0002 27-0103 27-0104 27-0117 27-0018 28-0101 28-0102 28-0103 28-0104 28-0117 28-0018 29-0101 29-0102 29-0103 29-0104 29-0117 29-0018
II	27-0002 27-0117 28-0102 28-0117 29-0102 29-0117	27-0002 27-0117 28-0102 28-0117 29-0102 29-0117
III	27B0005 27B0006 27B0008 27B0015 27B0016 27B0020 28B0005 28B0006 28B0008 28B0015 28B0016 28B0020	27B0005 27B0006 27B0007 27B0008 27B0009 27B0015 27B0016 27B0019 27B0020 28B0005 28B0006 28B0007

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TABLE II. Group qualification - Continued.

Group	Submission and qualification of any of the following connectors M39012/	Qualifies the following connectors M39012/
III	29B0005 29B0006 29B0008 29B0015 29B0016 29B0020	28B0008 28B0009 28B0015 28B0016 28B0019 28B0020 29B0005 29B0006 29B0007 29B0008 29B0009 29B0015 29B0016 29B0019 29B0020
IV	27B0007 27B0009 27B0019 28B0007 28B0009 28B0019 29B0007 29B0009 29B0019	27B0007 27B0009 27B0019 28B0007 28B0009 28B0019 29B0007 29B0009 29B0019
V	27-0010 27-0011 27-0013 27-0022 27-0023 28-0010 28-0011 28-0013 28-0022 29-0010 29-0011 29-0013 29-0022	27-0010 27-0011 27-0012 27-0013 27-0023 27-0014 27-0021 27-0022 28-0010 28-0011 28-0012 28-0013 28-0014 28-0021 28-0022 29-0010 29-0011 29-0012 29-0013 29-0014 29-0021 29-0022
VI	27-0012 27-0014 27-0021 27-0023 28-0012 28-0014 28-0021 29-0012 29-0014 29-0021	27-0012 27-0014 27-0021 27-0023 28-0012 28-0014 28-0021 29-0012 29-0014 29-0021

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TABLE II. Group qualification - Continued.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
VII	27-0501 27-0503 27-0504	27-0501 27-0502 27-0503 27-0504
VIII	27-0502	27-0502

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Dash number 1/ 2/ 3/	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
M39012/27B0005	028-RG058*	A	1.500 (38.10)
M39012/27B0006	060-RG142@* 128-RG400		
M39012/27B0007	29-RG59*Δ 030-RG062Δ 97-RG210Δ		
M39012/27B0008	054-RG122*		
M39012/27B0009	90-RG71*Δ		
M39012/27B0015	111-RG303*@		
M39012/27B0016	084-RG223*		
M39012/27B0019	110-RG302@*Δ		
M39012/27B0020	113-RG316@* 119-RG174		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.
2/ For maintenance replacements for category B, see table V.
3/ Inactive for new design.
The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable except as in notes @ and Δ.
@ Cable to be used for the +200°C temperature cycling tests.
Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number M39012/27-	Substitute for part number or type designation 1/ 2/
B0005 3/	M39012/27-0005
B0006 3/	M39012/27-0006
B0007 3/	M39012/27-0007
B0008 3/	M39012/27-0008
B0009 3/	M39012/27-0009
B0015 3/	M39012/27-0015
B0016 3/	M39012/27-0016
B0019 3/	M39012/27-0019
B0020 3/	M39012/27-0020
0010	M23329/4-07, M23329/4-09
0011	M23329/4-08, M23329/4-10
0012	M23329/4-11
0023	M23329/4-12
0101	M39012/27-0001
0103	M39012/27-0003
0104	M39012/27-0004
0117	M39012/27-0017

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/27-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new 'B' part numbers will be required marking 6 months from the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number* Inactive for new design	Category C dash number	Category A dash number	Category D dash number
B0005	0010	0101	0504
B0006	0011	0101	0503
B0007	0012	0002	---
B0008	0013	0103	0501
B0009	0014	0002	---
B0015	0010	0104	---
B0016	0011	0101	---
B0019	0021	0117	---
B0020	0022	0018	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the change.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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